Parametric Explorer Guide

LENR Sonoluminescence Reactor

Interactive Cavity Acoustics Tool

Contributor: dbRollin

Repository: github.com/JKocher/LENR

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1. Purpose

The Parametric Explorer provides real-time visualization of how acoustic resonances in a spherical cavity respond to changes in geometry, fluid properties, and wall material. It replaces the need to run a full Elmer FEM solve for each configuration by using validated analytical models that update instantly as parameters change.

The tool opens at the reactor design point specified in the schematic (v0.1.0): 160C operating temperature, 10% D2O in H2O, aluminum acoustic horns. The reactor operates pressurized (200 PSI nominal, 300 PSI blowoff) to maintain liquid water above 100C. Temperature range extends to 300C using IAPWS-95 data for pressurized water.

The tool is designed to answer three practical questions:

1. Which transducer frequency best matches a given cavity design?
2. What does the internal pressure field look like at resonance?
3. How sensitive are the resonances to temperature, fluid chemistry, and wall material?

To launch: cd "Simple Python Sim" && python parametric_explorer.py
Requires: numpy, matplotlib (no scipy dependency).

2. Interface Overview

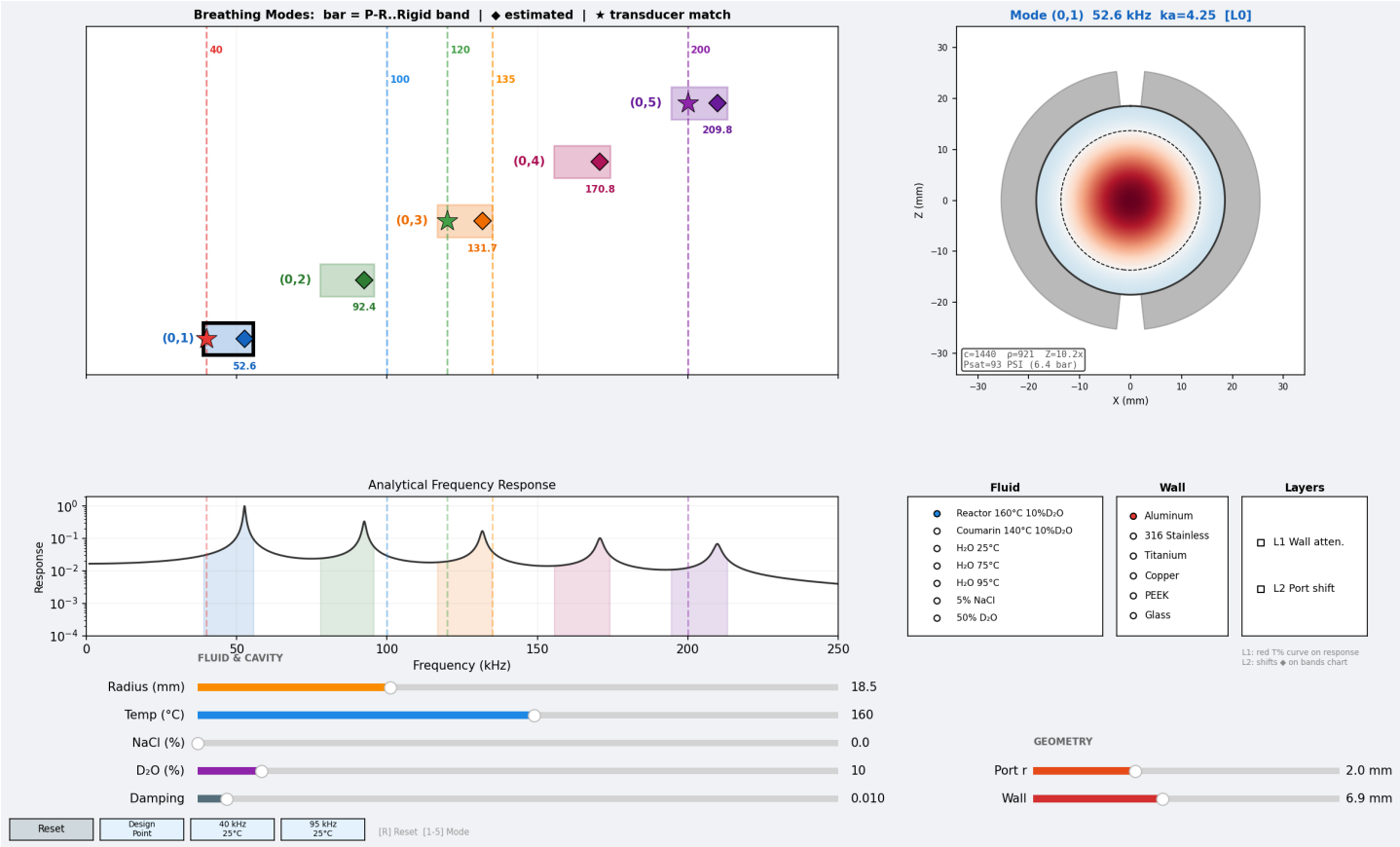

Figure 1: Explorer at reactor design point (160C, 10% D2O, aluminum)

The interface has four main areas: the breathing modes chart (top-left), the cross-section view (top-right), the frequency response plot (middle-left), and the controls panel (presets, layers, and sliders along the bottom and right). All charts update simultaneously when any parameter changes.

3. Breathing Modes Chart

The top-left chart is the primary decision tool. It shows the frequency range where each breathing mode can resonate, overlaid with available transducer frequencies.

Frequency bands (horizontal bars)

Each colored bar spans a frequency range bounded by two theoretical limits. The left edge is the pressure-release (P-R) limit, which assumes the cavity wall is acoustically transparent. The right edge is the rigid-wall limit, which assumes perfect reflection. Any real wall material produces a resonance somewhere between these extremes. The width of the bar represents the range of uncertainty due to boundary conditions.

Estimated frequency (diamond markers)

The diamond on each bar marks the estimated resonant frequency for the current wall material. Its position depends on the acoustic impedance ratio Z_wall / Z_fluid. Softer walls (lower impedance ratio) place the diamond closer to the P-R edge; harder walls push it toward the rigid edge. The frequency value in kHz is printed below each marker.

Transducer overlay (dashed lines and stars)

Dashed vertical lines represent common off-the-shelf transducer frequencies (40, 100, 120, 135, 200 kHz). When a transducer line falls within a mode band, a star marker appears at the intersection, indicating that transducer could potentially excite that mode.

Mode notation

Mode labels use standard spherical harmonic notation (n, m). All modes shown are breathing modes (n = 0), meaning they are radially symmetric with no angular variation. The index m indicates radial order: mode (0,1) has a single pressure zone, mode (0,2) has two concentric zones separated by a node ring, and so on.

4. Wall Material Effect

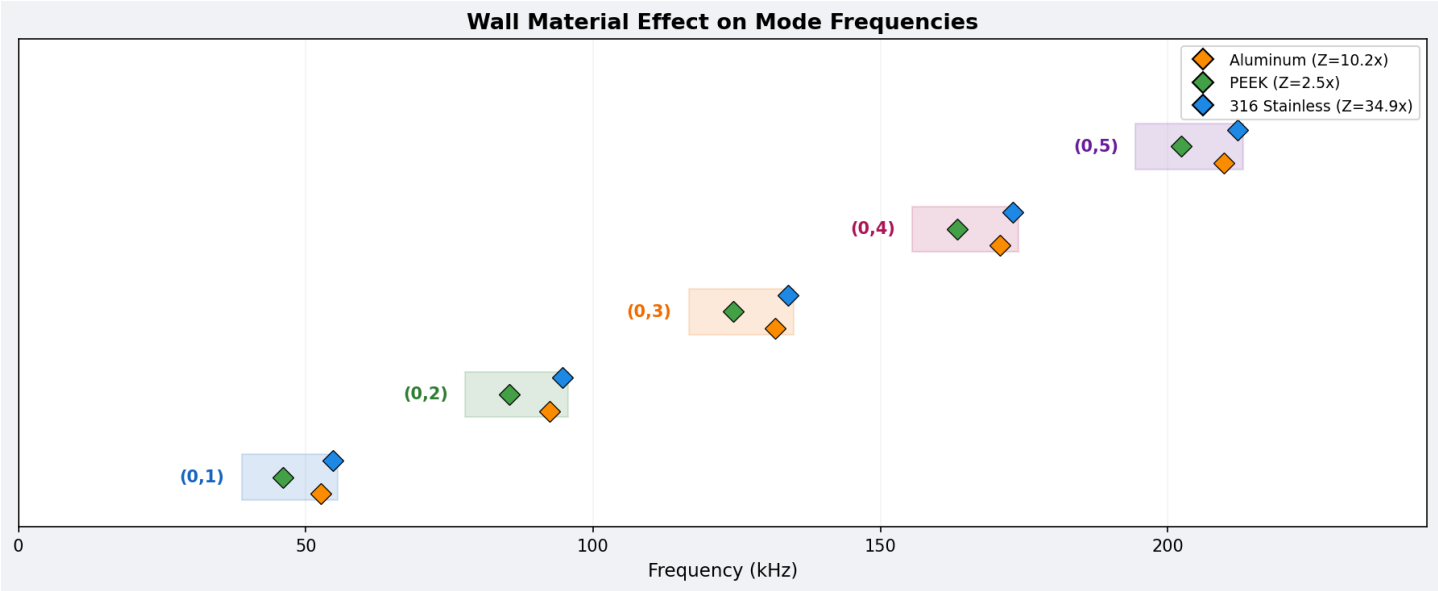

Figure 2: Diamond positions shift with wall material

The wall material determines where each resonance falls within its band. Higher impedance ratio (harder wall relative to the fluid) shifts the estimated frequency upward. For example, PEEK (a polymer, Z = 2.2x) produces frequencies near the P-R limit, while 316 stainless steel (Z = 31x) pushes them toward the rigid limit. The current design uses aluminum horns (Z = 9.0x).

Available wall presets with (sound speed, density): Aluminum (5000 m/s, 2700 kg/m3), 316 Stainless (5790, 8000), Titanium (6070, 4500), Copper (4760, 8960), PEEK (2530, 1300), Glass (5640, 2230).

5. Cross-Section View

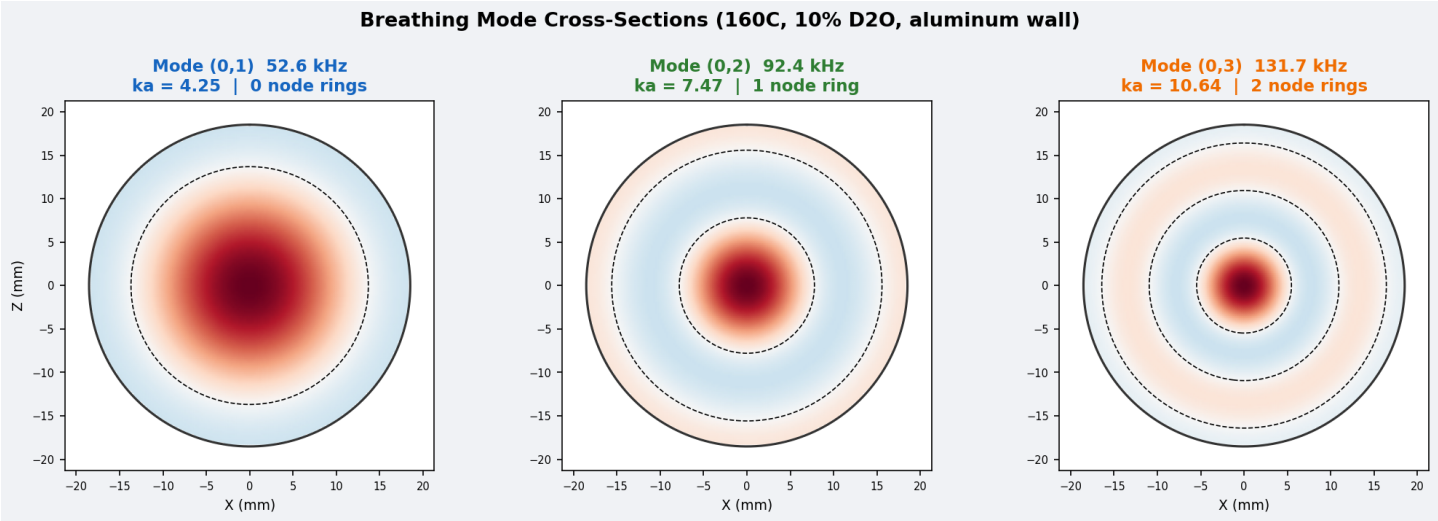

Figure 3: Modes (0,1) through (0,3) showing increasing node rings

The top-right panel shows a diametric slice through the cavity, displaying the predicted pressure field for the selected breathing mode. Color indicates pressure amplitude: red for compression (positive), blue for rarefaction (negative), and white for zero pressure (nodes). The center of a breathing mode is always a pressure maximum -- this is where acoustic energy concentrates.

Dashed black circles mark node rings -- radial positions where the pressure passes through zero. Mode (0,1) has no internal nodes (uniform positive pressure throughout). Mode (0,2) has one node ring at approximately half the cavity radius. Mode (0,3) has two. These contours are computed directly from the spherical Bessel function j_0(kr) and represent real physical predictions.

The grey annular ring represents the cavity wall. Gaps at the poles correspond to utility openings in the hemisphere (used for water circulation or observation windows). Click a mode bar or press [1]-[5] to change the displayed mode.

6. Frequency Response

The middle-left chart shows the analytical frequency response of the cavity -- equivalent to sweeping a signal generator across all frequencies and measuring the resulting pressure amplitude. Peaks in this curve correspond to resonant modes. Peak height and sharpness are governed by the Damping parameter: low damping produces tall, narrow peaks (low-loss cavity); higher damping broadens the peaks (more realistic).

Colored shading under the curve identifies which frequency ranges correspond to each breathing mode. Transducer reference frequencies appear as dashed vertical lines, consistent with the bands chart above.

7. Analysis Layers

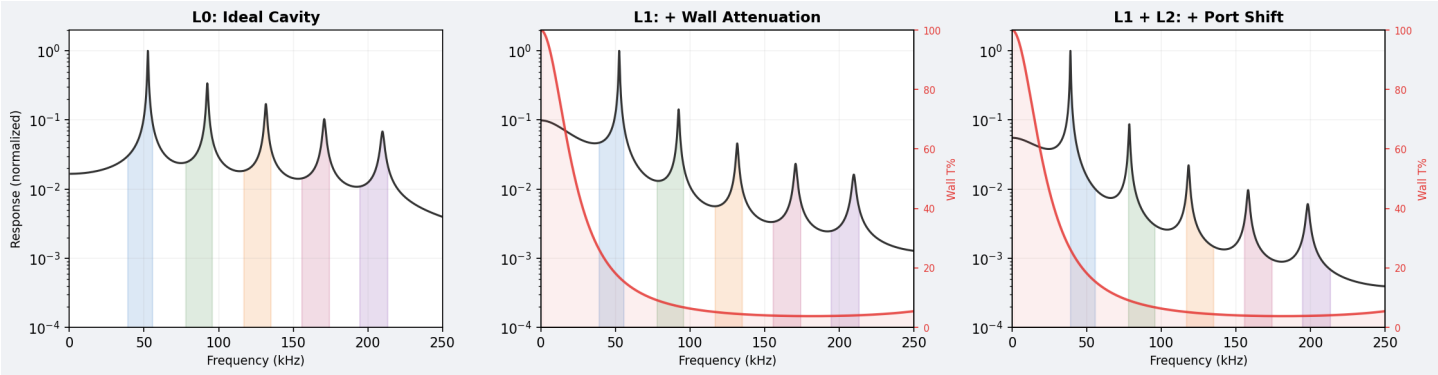

Figure 4: Progressive refinement -- L0, L1, and L1+L2

L0: Ideal Cavity (default)

The base model treats the cavity as a perfect sphere with uniform boundary conditions. Resonant frequencies are determined solely by the acoustic impedance ratio between the wall material and the fluid. This is what the explorer shows with both layer checkboxes off.

L1: Wall Attenuation

Enables a wall transmission model based on the transfer matrix method for a planar slab. In a real system, acoustic energy must propagate through the wall to enter the cavity. The wall acts as a frequency-dependent filter: at certain frequencies it transmits efficiently, while at others it reflects most of the energy. When L1 is active, a red transmission curve (T%) appears on the response plot, and the response peaks are attenuated accordingly. This can reveal that some modes are effectively blocked by the wall geometry.

L2: Port Correction

Applies a frequency-dependent boundary condition correction for utility openings (ports) in the cavity wall. These openings -- used for water circulation or observation -- create local pressure-release boundaries because the fluid is exposed to the outside rather than being bounded by a rigid wall.

The correction is frequency-dependent: at low frequencies, where the acoustic wavelength is much larger than the port diameter, the open area has a strong influence and pulls the effective boundary condition toward pressure-release. At high frequencies, where the wavelength is small relative to the port, the wall impedance dominates. When L2 is active, the diamond markers on the bands chart shift leftward (especially for lower-frequency modes), and the cross-section updates accordingly.

Note on L1/L2 maturity

The L1 and L2 layers are exploratory refinements. The underlying physics models (transfer matrix wall transmission, frequency-dependent port BC blending) are well-established, but the specific parameterizations may need further tuning against experimental or FEM data to be fully quantitative. They are disabled by default and provided as a starting point for future investigation. The L0 base model is the primary analysis tool.

8. Controls Reference

Fluid & Cavity sliders (bottom-left)

* Radius (mm): Inner radius of the spherical cavity. Default 18.517 mm matches the upstream CAD geometry. Changing radius shifts all mode frequencies (smaller cavity = higher frequencies).

* Temp (C): Water temperature, 5-300C. Uses Marczak (1997) polynomial below 95C and IAPWS-95 lookup table above. Sound speed peaks near 75C, so mode frequencies shift non-linearly with temperature. Above 100C, saturation pressure is shown on the cross-section panel.

* NaCl (%): Dissolved salt concentration, 0-10% by weight. Increases sound speed (~14 m/s per 1%) and density. Based on the Chen-Millero correlation.

* D2O (%): Heavy water fraction, 0-100%. D2O has lower sound speed (1400 vs 1497 m/s at 25C) and higher density (1104 vs 997 kg/m3). Mixing uses bulk modulus blending for physical consistency. Based on Herrig (2018) tabulated data.

* Damping: Acoustic loss factor. Controls resonance peak width, not frequency. Low values (0.001) represent an idealized cavity; higher values (0.05+) approximate real-world losses.

Geometry sliders (bottom-right)

* Wall (mm): Wall thickness, 1-15 mm. Affects the L1 wall transmission filter characteristics and the grey annulus in the cross-section. Default 6.9 mm (from upstream geometry).

* Port r (mm): Utility port opening radius, 1-15 mm. Affects the L2 port correction strength and the gap shown in the cross-section wall. Default 9.5 mm (from upstream geometry).

Presets

* Fluid presets: Load fluid configurations ranging from the reactor design point (160C, 10% D2O) to benchtop baselines (25C water). Sets the three fluid sliders simultaneously.

* Wall presets: Select a wall material. Each preset sets sound speed and density as a coupled pair -- these are physical properties of the material and cannot be adjusted independently.

Target Scenarios

* Design Point: Loads the reactor nominal conditions -- 160C, 10% D2O, aluminum walls, mode (0,1). This is also the Reset state.

* 40 kHz 25C: Benchtop baseline with mode (0,1) selected. Room-temperature water for comparison testing.

* 95 kHz 25C: Benchtop baseline with mode (0,2) selected. Targets the second breathing mode which produces strong center pressure buildup (confirmed by FEM).

Interaction

* Click a mode bar in the bands chart to select it for cross-section display.

* Press [1]-[5] to select a mode by number.

* Press [R] or click the Reset button to restore default parameters.

* All three charts update in real time when any control changes.

9. Target Configurations

Three target configurations are built into the explorer as quick-access buttons. All use the reference cavity geometry (18.517 mm radius, aluminum walls). The Design Point button matches the reactor schematic (v0.1.0) nominal conditions: 160C, 10% D2O, 200 PSI.

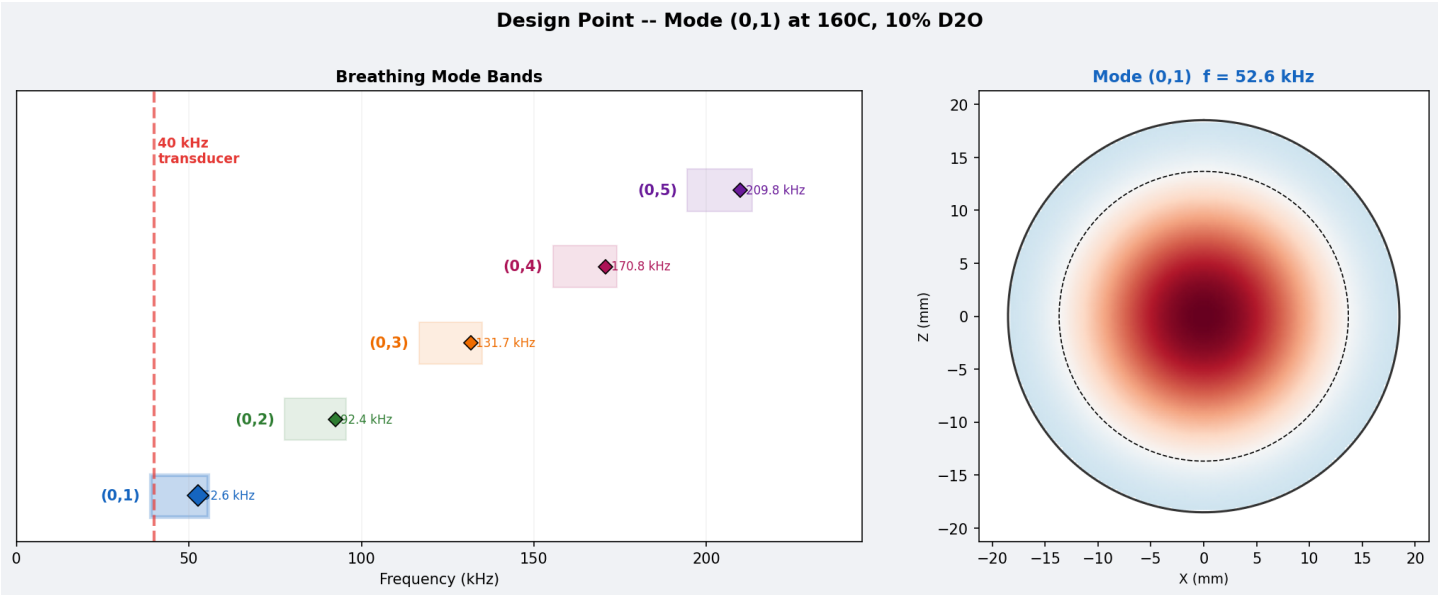

Figure 5: Design Point -- mode (0,1) at 160C, 10% D2O

Design Point -- Reactor Nominal

At the reactor operating temperature (160C) with 10% D2O, the sound speed drops to ~1440 m/s and density to ~921 kg/m3. This shifts mode (0,1) down to 38.9-55.6 kHz. The saturation pressure at 160C is approximately 90 PSI (6.2 bar), well within the reactor's 200 PSI design limit. The schematic specifies operating at the system's resonant frequency for maximum Bremsstrahlung output.

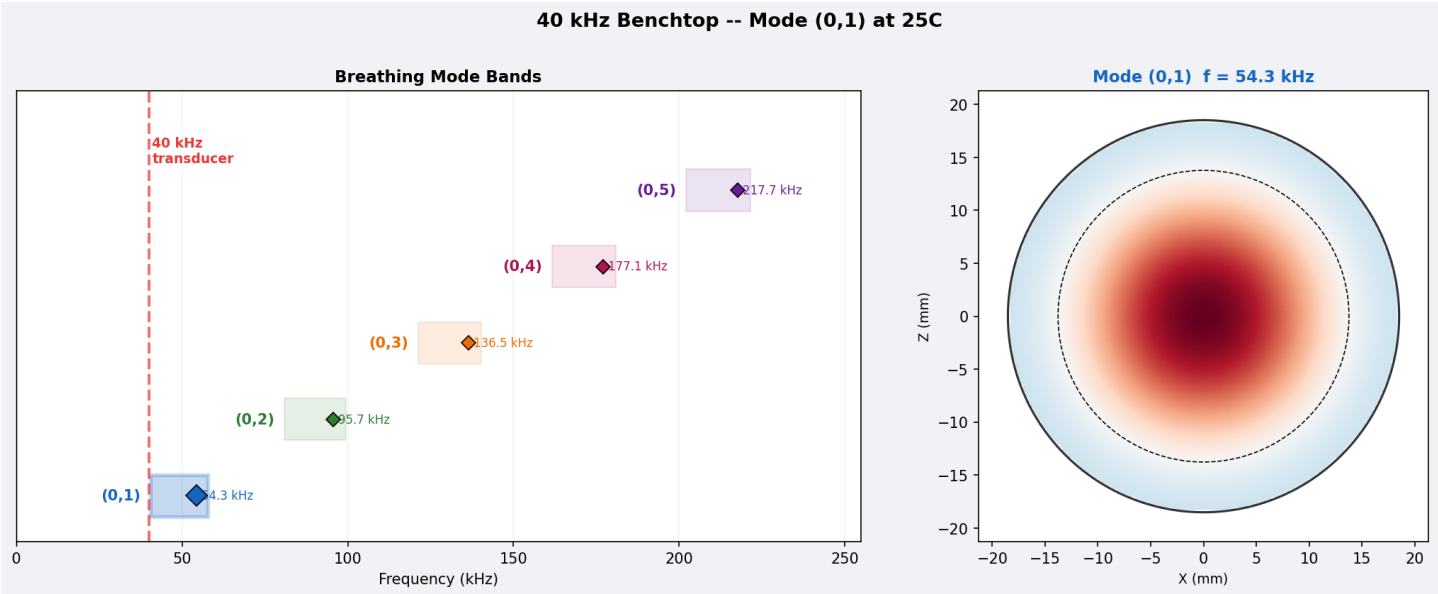

Figure 6: 40 kHz benchtop -- mode (0,1) at 25C

40 kHz Benchtop -- Mode (0,1)

At room temperature (25C), mode (0,1) spans 40.4-57.8 kHz. A standard 40 kHz transducer sits at the pressure-release edge. This configuration is useful for benchtop validation before pressurized operation. Elmer FEM confirms a resonance peak at 41 kHz.

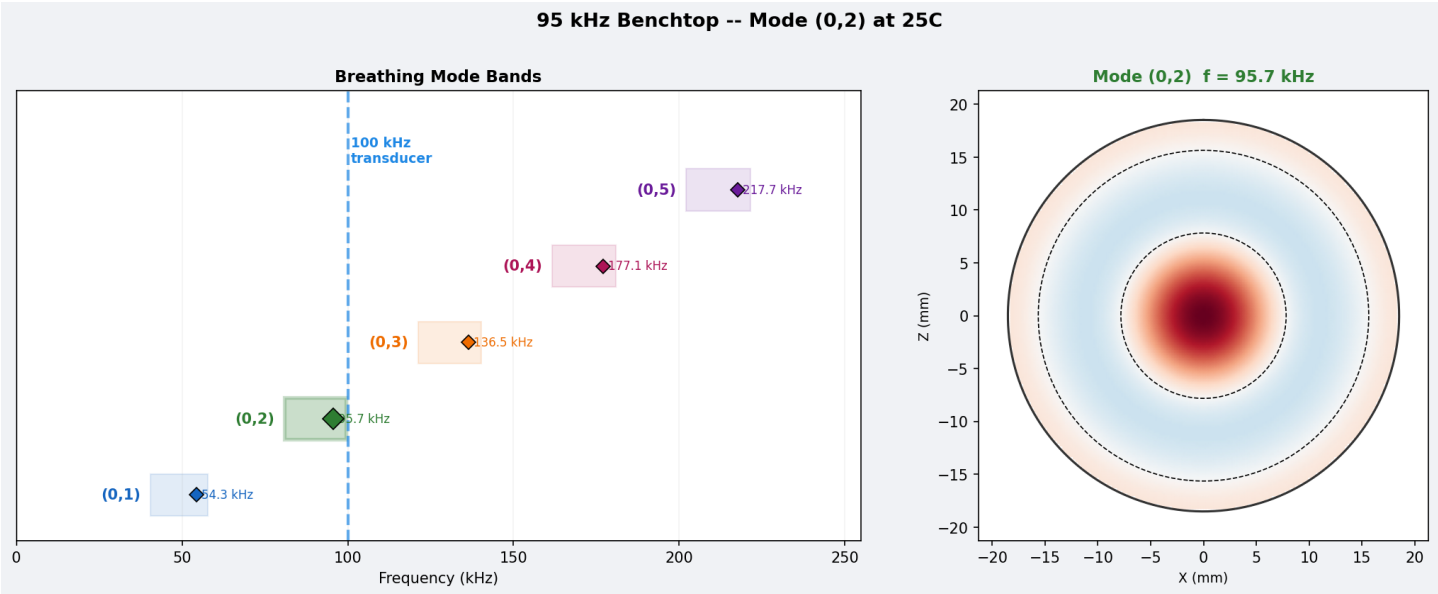

Figure 7: 95 kHz benchtop -- mode (0,2) at 25C

95 kHz Benchtop -- Mode (0,2)

The second breathing mode (0,2) at 25C spans 80.8-99.4 kHz with estimated resonance at ~95.7 kHz. Unlike mode (0,1), this mode produces strong center pressure (FEM center amplitude ~110 at 95 kHz), making it especially relevant for sonoluminescence research. A 100 kHz transducer is the closest standard frequency.

10. FEM Cross-Validation

The analytical predictions were cross-validated against Elmer FEM frequency sweeps using corrected material units (mm-tonne-s system to match the mm-scale mesh). The FEM solves the full 3D Helmholtz equation on the actual reactor geometry including horn walls and port openings.

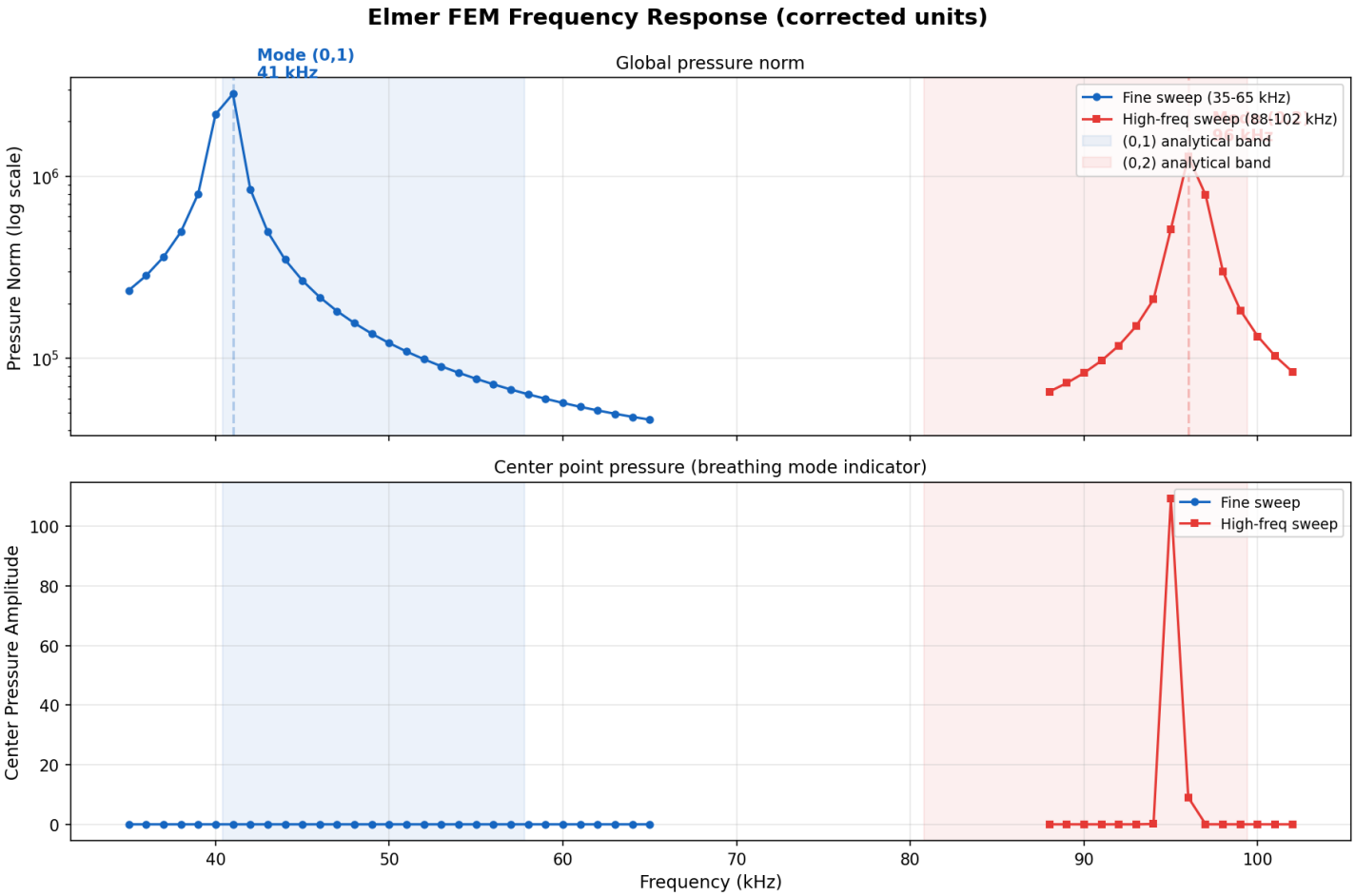

Figure 8: Elmer FEM frequency response (corrected units)

Top panel: global pressure norm (log scale) shows resonance peaks at 41 kHz (mode 0,1) and 96 kHz (mode 0,2). Shaded bands show the analytical predictions -- both FEM peaks fall within the predicted ranges. Bottom panel: center pressure amplitude identifies the true breathing mode at 95 kHz where pressure concentrates at the cavity center, confirming the analytical model's prediction of 95.7 kHz.

11. Validation

All physics correlations are verified by an automated test suite (validate_acoustic_model.py) that runs 90 checks against published reference data:

* Level 1 -- Correlation accuracy: 15 checks against Marczak (1997), Kell (1975), Herrig (2018), Chen-Millero, and IAPWS-95 reference tables. Tolerances derived from published error bounds.
* Level 2 -- Parameter bounds: 51 checks verifying all fluid/wall combinations (including reactor design point at 160C, 10% D2O) produce physically reasonable values (sound speed 900-1600 m/s, density 700-1200 kg/m3, all impedance ratios > 1).
* Level 3 -- Mode predictions: 10 checks confirming breathing mode frequencies at both 25C reference and 160C reactor design point match known analytical solutions.
* Level 4 -- Physics sanity: 14 checks on edge cases and mathematical identities (zero-thickness wall = full transmission, matched impedance = pressure-release limit, Bessel zero constants, field symmetry).
* Level 5 -- FEM cross-validation (optional): Compares analytical predictions against Elmer FEM sweep results when available.

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To run:  python validate_acoustic_model.py        (Levels 1-4)
         python validate_acoustic_model.py --fem  (Include Level 5)
```

12. File Architecture

The explorer consists of three files with clear separation of concerns:

* acoustic_model.py -- Shared physics library containing fluid property correlations, wall material presets, breathing mode computation, wall transmission (transfer matrix method), port boundary condition correction, and pressure field calculation. Imported by both the explorer and cavity_modes.py. Requires only numpy.
* parametric_explorer.py -- Interactive GUI built on matplotlib. Imports all physics from acoustic_model.py and provides sliders, presets, and real-time visualization. Requires numpy and matplotlib.
* validate_acoustic_model.py -- Automated test suite. Verifies every correlation in acoustic_model.py against published reference data. Requires only numpy.

The existing cavity_modes.py (unchanged by this contribution) can also import from acoustic_model.py when scipy is available, enabling exact Bessel zero computation. The explorer avoids the scipy dependency by using precomputed zeros.